

AI-Powered Fire Prediction Spread and Containment System

User Manual

Panic! At The Kernel
Client: EPI-USE Africa

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FIRE**AWAY**

AI-POWERED FIRE SPREAD PREDICTION AND CONTAINMENT SYSTEM



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1 Introduction

Fire Away is an AI-powered fire spread prediction system that uses real-time weather data, environmental conditions, and user reported fire locations to simulate fire progression and support emergency responders with early warnings and realtime strategic containment guidance.

2 Getting Started

2.1 Key features

- **Live Fire Map View:** View active fires near you based on your current location.
- **Reporting an Active Fire:** Report an active fire at a specific location including an image upload of the fire.
- **Firefighter Dashboard:** Aids firefighters in keeping track of changing weather and environmental conditions as well as active fires.
- **Interactive AI-Powered Fire Spread Simulation:** With the help of our AI model, watch how a fire can spread based on environmental factors and test how the fire can be contained.
- **Admin Analytics Dashboard:** Keep track of how many users are registered on the app and approvals.

2.2 Accessing Fire Away

Fire away is currently accessible at <https://fireaway.csml.co.za/> on your desktop browser

3 User Roles and Responsibilities

3.1 Admin

Administrators oversee the system, user access, fire report record-keeping and manual verification and rejection. Their responsibilities include:

- Approving or rejecting registration requests for the Administrator role.
- Managing user roles: approving/rejecting role requests (for admin and firefighters), searching users, and revoking access.
- Maintaining the searchable incident archive, into which closed fire incidents are automatically moved.
- Manually verifying or rejecting pending fire reports.

3.2 Firefighter

Firefighters are the field responders, using tactical tools for fire containment efforts and inform the AI simulation. Their responsibilities include:

- Monitoring the tactical dashboard, which displays fire direction, high-risk areas, wind conditions, and AI-recommended containment locations.
- Logging containment lines on the live map, triggering an automatic simulation re-run to reflect field actions.

- Operating with a live GPS position indicator on the map during active incidents.
- Accessing cached map data and the most recent spread prediction while offline, to maintain situational awareness in low-connectivity areas.
- Manually verifying or rejecting pending fire reports.

3.3 Registered User

Registered Users are members of the public who have created an account to receive awareness. Their responsibilities include:

- Submitting fire reports and will receive status updates on their own reports.
- Receiving proximity-based push notifications when a fire is in close proximity to them.
- Viewing the AI-generated spread-prediction overlay on the live map, which is not available to guests.

3.4 Guest

Guests (Anonymous visitors) interact with the system without creating an account. Their responsibilities include:

- Viewing the live fire map (including verified fire markers) without logging in.
- Submitting a fire report without an account.

4 Account Management

4.1 Sign-in as a Guest

Visitors who do not want to create an account can sign in as a guest user immediately.

- Navigate to the Fire Away login page.
- Select the '**Sign in as Guest**' button beneath the Login and Register options.
- From there, you will be taken directly to the Guest Dashboard, where you can view the live public fire map without needing to provide any credentials.

As a Guest, no personal information is collected and no session cookie tied to an account is created. You retain full access to the live fire map and fire reporting tools described in Section 5, but you will not receive proximity notifications.

4.2 Registering

To create an account, navigate to the login page and select **“Register.”** The registration form requires the following information:

- **Name and Surname**
- **Email address** – must be a valid email format
- **ID/Passport Number** – a 13-digit South African ID number
- **Password** – at least 8 characters, including at least one uppercase letter and one number

- **Confirm Password** – must match the password entered above
- **Role** – choose between *User* and *Firefighter*

If you select the *Firefighter* role, an additional **Licence Number** field appears and is required to complete registration. This allows Fire Away to distinguish verified firefighting personnel, who are granted access to the tactical dashboard described in later sections.

All fields are validated as you type, and any errors (such as a mismatched password or an invalid ID number) are shown directly beneath the relevant field before the form can be submitted.

Once the form is submitted successfully, you will be redirected to a two-factor authentication (2FA) setup page, where you can scan a QR code to link an authenticator app to your new account. If registration fails – for example, because the email address is already in use – an error message is displayed at the bottom of the form and no account is created.

4.3 Logging in

Returning users can log in from the main Fire Away login page by entering their registered **email** and **password**. Both fields are validated before submission: the email must be correctly formatted, and the password must be at least 6 characters long.

Once your credentials are submitted:

- If two-factor authentication is required on your account, you will be redirected to a verification page to enter your one-time passcode.
- If login is successful, you are automatically redirected to the dashboard matching your account role – the Admin dashboard, the Firefighter dashboard, or the Registered User dashboard.
- If the email or password is incorrect, an error message is displayed below the form and you remain on the login page.

If you have forgotten your password or are otherwise unable to log in, the “**Forgot password?**” and “**Can’t log in?**” links beneath the login form will guide you through account recovery.

5 Live Fire Map

The Live Fire Map is the central public safety feature of Fire Away and is accessible to Guests and Registered Users alike, displayed under the heading “**Incident Map.**”

When the page loads, Fire Away requests permission to use your device’s location. If permission is granted, the map centers on your current position; otherwise, it falls back to a default location. The map itself shows active and verified fires in your surrounding area, refreshed automatically as new data becomes available.

Alongside the map, the dashboard displays two additional panels:

- **Environmental Conditions** – current weather and environmental readings (such as wind and temperature) for your area, used as inputs to the fire-spread predictions.
- **Nearby Reports** – a scrollable list of fire reports submitted within a 20 km radius of your location, giving a quick overview of recent activity near you.

A **Recenter** action is available at any time, which re-requests your device’s location and re-centers the map and nearby report list around your current position. If location access is unavailable, the map recenters on the system default location instead.

From the sidebar, you can also navigate directly to the **Report Fire** page to submit a new fire report (see Section ??).

6 Reporting a Fire

All the roles have access to the Reporting a Fire page. The users are able to report a fire and add evidence of the fire to create a fire report. When it is submitted, the status is set to pending, and the fire report needs to go through the verification process, which is the verification algorithm, or it gets manually verified or rejected by an admin or firefighter. When it is verified, the report status is set to verified, and alerts will be sent out to the registered users in close proximity to the fire, and the fire map will be updated accordingly. The fire report's status is visible to the user on the report a fire page.

REPORT A FIRE

1 Drop a pin on the map 2 Drag boundary ring to show size 3 Add details and submit

LIVE MAP

Report details

Location

106 Jeppe Street, Johannesburg, Gauteng 2001, South Africa

Description optional

E.g., Surface line fire spreading northeast toward residential properties...

Attach Evidence

Attach Image

SUBMIT FIRE REPORT

Report status

Verified #FR-2026-05E878
Push notifications sent out

6.1 Reporting a fire

1. Click a location on the map or search the location in the search bar.
2. Adjust the circle to the approximate size of the fire.
3. The user has the option to add a description.
4. Upload a picture of the fire for evidence.
5. Click on the submit button to submit the fire report.
6. The fire report status changes visibly.

7 Nearby Fires

The Nearby Fires Panel is displayed on every user role's landing page/dashboard. This displays a list of active fires within a 20km radius of your current location.

NEARBY REPORTS		
LC de Villiers Sports Grounds, Hatfield	1.4 km · 2 days ago	verified >
Lynnwood Road crossing	2.5 km · 2 days ago	pending >
Pretoria National Botanical Garden	4.5 km · 2 days ago	verified >
Groenkloof Nature Reserve	4.6 km · 2 days ago	pending >
Fountains Valley Recreation Resort	5 km · 2 days ago	pending >
Menlyn Maine construction site brush	6.2 km · 2 days ago	pending >
Voortrekker Monument hillside	6.4 km · 2 days ago	verified >
Struben Dam Bird Sanctuary	6.8 km · 2 days ago	pending >
Faerie Glen Nature Reserve	7.1 km · 2 days ago	verified >
Vacant lot, Erasmuskloof	7.5 km · 2 days ago	verified >
Pretoria West Industrial Area	8 km · 2 days ago	verified >

8 Reported Fires (Firefighter & Admin)

The Reported fires page shows all the reported fires in a table view. Admins and firefighters have access to this page. The reported fires can be filtered to All, Pending, Verified and Rejected by clicking on the labels. A specific reported fire can be searched for by location, reference, reporter etc. When clicking on the View button, a pop-up is shown with more details of the fire, including an image that the reporter added if he or she were so inclined. An admin or firefighter can then manually approve or reject a pending fire and they can revoke an approved status from a reported fire that has already been approved.

8.1 Managing a fire report

1. Open the Reported Fires page to view the table of all submitted fire reports.
2. Click the **View** button on a report to open a pop up with more details, including the fire's location on the map and any image the reporter added as evidence.
3. If the report is **Pending**, click **Verify** to manually approve it or **Reject** to decline it.
4. If the report is already **Verified**, click **Revoke** to undo the approval if the report was falsely verified.
5. If the report is **Rejected**, click **Re-verify** to send it back for review.
6. The report's status updates visibly once an action is taken.

REPORTED FIRES

Manage and review fire reports

All

Pending

Verified

Rejected

Q

Search by location, ref, rep

REF	LOCATION	STATUS	SIZE	REPORTED	REPORTER	VIEW
FR-2026-CBC7CF	17 Tudhope Avenue, Johannesburg, Gauteng 2198, South Africa	Verified	12.6 ha	09 Aug 02:30	Amahle Dlamini	View
FR-2026-44016B	M27, Johannesburg, Gauteng 2001, South Africa	Rejected	12.6 ha	09 Aug 02:26	Sipho Ndlovu	View
FR-2026-004	Rietvlei Nature Reserve	Verified	3848.5 ha	06 Aug 21:54	Heinrich Kruger	View
FR-2026-005	Moreleta Kloof Nature Reserve	Pending	314.2 ha	06 Aug 21:54	Anonymous	View
FR-2026-006	Faerie Glen Nature Reserve	Verified	706.9 ha	06 Aug 21:54	Zanele Mbatha	View
FR-2026-007	Wonderboom Nature Reserve	Pending	1963.5 ha	06 Aug 21:54	Ruan Venter	View
FR-2026-008	Pretoria National Botanical Garden	Verified	28.3 ha	06 Aug 21:54	Naledi Moeng	View
FR-2026-009	Roodeplaat Dam Nature Reserve	Verified	5026.5 ha	06 Aug 21:54	Anonymous	View

Fire Report Location

Photo Evidence

Report Details

Reference

FR-2026-44016B

Status

Pending

Reporter

Sipho Ndlovu

Location

M27, Johannesburg, Gauteng 2001, South Africa

Size

12.6 ha

Reported at

09 Aug | 02:26

Description

No description provided.

Action

Review the fire report. Reject or verify manually.

Verify

Reject

9 Fire Spread Simulation

The interactive Fire Spread Simulation page is used by firefighters. The simulation allows for firefighters to visualize how a fire spread of a certain period of time based on live weather and environmental conditions. Firefighters are also able to draw containment lines on the map to visualize where to place containment systems to stop/slow the spread of fire.

9.1 Running the Interactive Fire Spread Simulation

Using Auto Play

1. Ensure the checkbox labeled 'Auto Play', has been checked

8

Fire Report Location

Photo Evidence

Report Details

Reference	FR-2026-440168
Status	Verified
Reporter	Sipho Ndlovu
Location	M27, Johannesburg, Gauteng 2001, South Africa
Size	12.6 ha
Reported at	09 Aug 02:26

Description

No description provided.

Action

This report has already been verified. Revoke if report is falsely verified.

Revoke

Fire Report Location

Photo Evidence

Report Details

Reference	FR-2026-440168
Status	Rejected
Reporter	Sipho Ndlovu
Location	M27, Johannesburg, Gauteng 2001, South Africa
Size	12.6 ha
Reported at	09 Aug 02:26

Description

No description provided.

Action

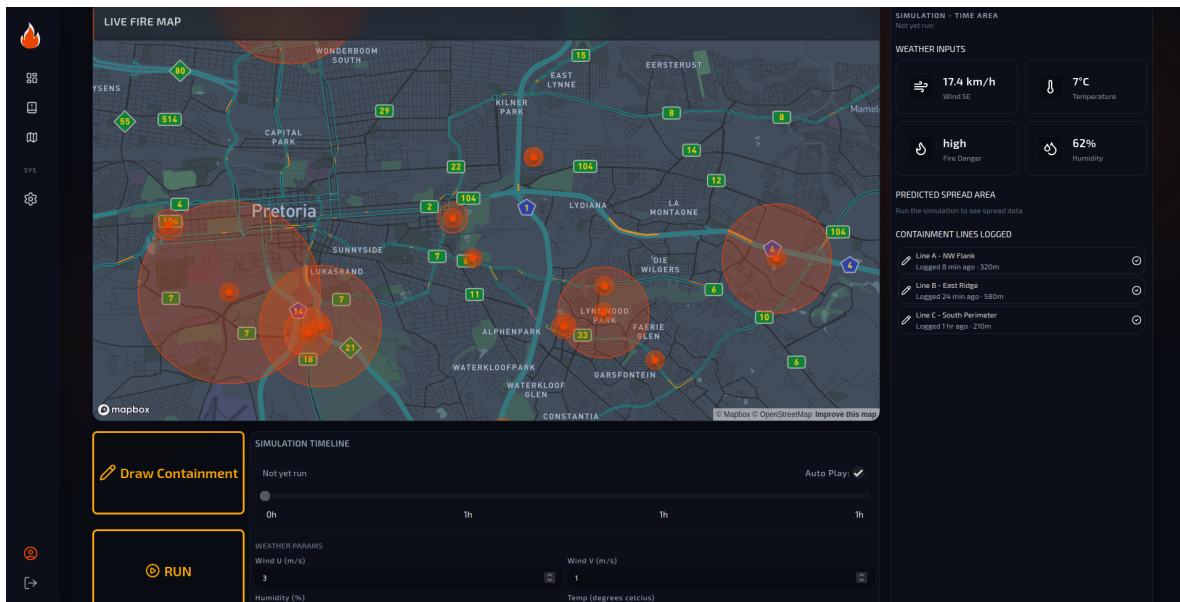
This report was rejected. Send to be re-verified.

Re-verify

2. Press the 'Run' button

Not Using Auto Play

1. Ensure the checkbox labeled 'Auto Play', has been unchecked
2. Press the 'Run' button
3. Slide the slider in the slider bar in order to view individual ticks at a specific point in time



9.2 Drawing/Logging Fire Spread Containment Lines

1. Click the 'Draw Containment' button
2. Click your desired location on the map where you want to place a containment line
3. Move your pointer to your desired destination location and double click to place the containment line

10 Firefighter Tactical Dashboard

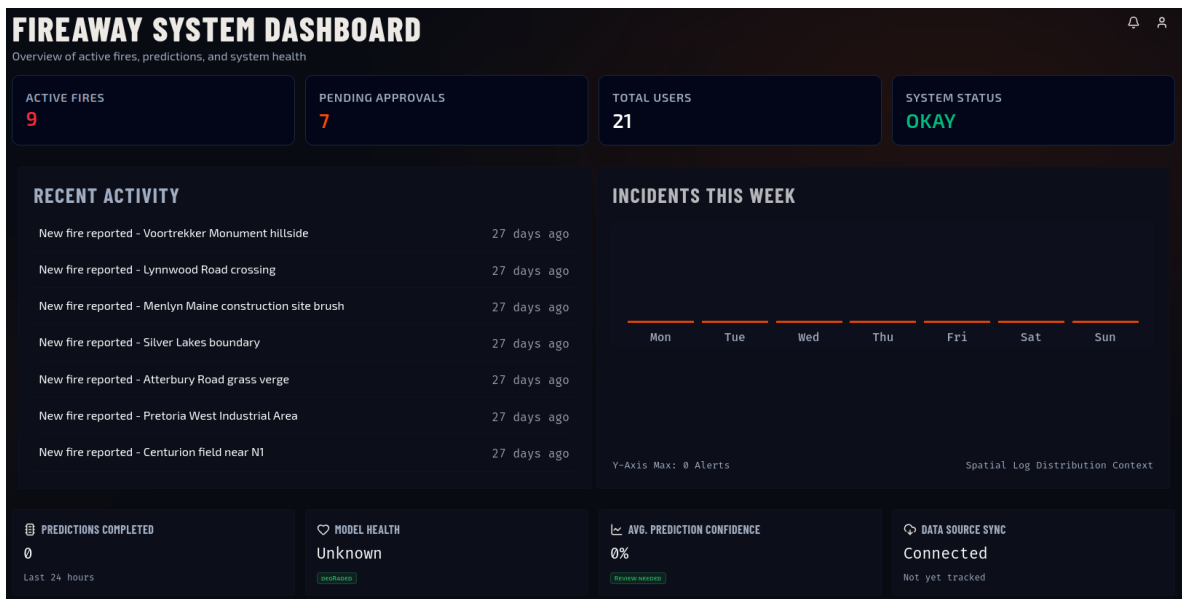
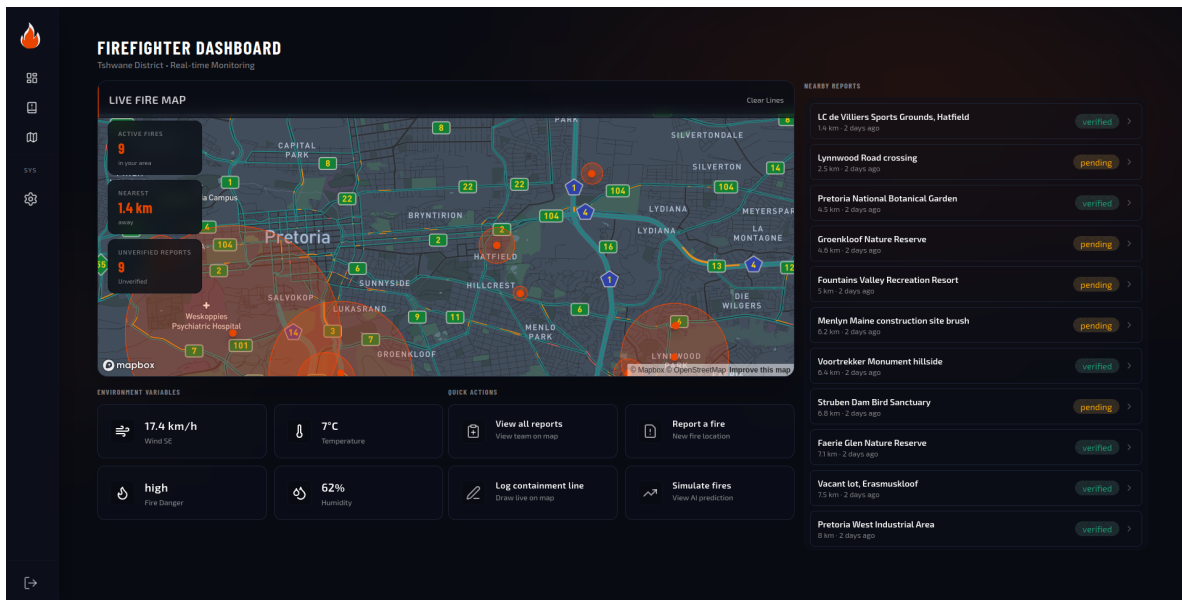
The Firefighter Tactical Dashboard is a page specifically designed to equip firefighters with all the relevant information they need to see as soon as they login. This page comprises of a live fire map, current weather and environmental conditions as well as nearby fires.

11 Admin Dashboard

The Admin Dashboard is accessible to admins only. Here they can see the total active fires, the amount of pending approvals, the total users of the web app and the system's status. There is a table of recent activity in terms of fires reported, role requests, role revocations and fire approvals and rejections. There is also a bar graph of the weekdays showing the incidents this week in terms of reported and verified fires. Each day is represented by a bar and when hovering over a bar an admin will also be able to see the exact number of incidents of each day. Further the model's metrics are displayed at the bottom. The number of predictions that have been completed, the model's health, the average prediction confidence and if the data sources are synced.

12 Admin Analytics

The admin analytics dashboard shows the total users, the number of pending role requests, the total firefighters and the total number of admins. Then it also shows the pending role requests.



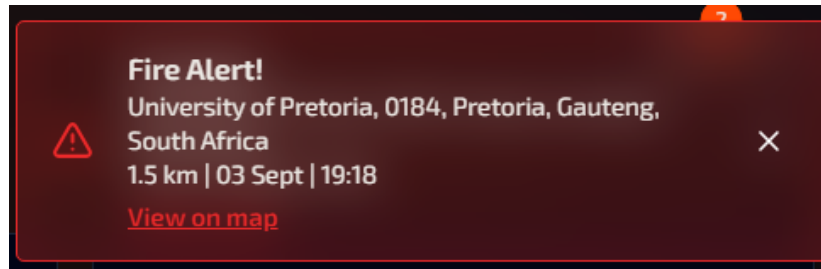
13 Notifications

FireAway keeps Registered Users informed of fire activity near them without requiring them to keep the app open. Notifications are only available to Registered Users. Guests do not receive alerts, since no account or location subscription exists to notify.

13.1 Proximity-Based Alerts

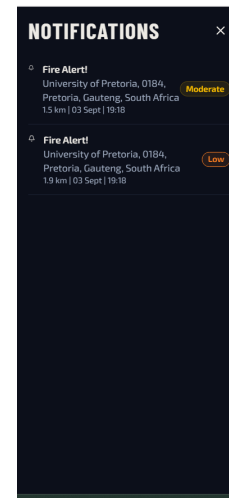
When a fire report is verified (either automatically or manually by an Admin or Firefighter), FireAway checks which Registered Users are located near the fire and sends each of them a push notification. This ensures that residents in the affected area are made aware of a nearby fire as soon as it is confirmed, rather than having to actively check the Live Fire Map.

1. A fire report is submitted and moves through the verification process.
2. Once the report's status changes to Verified, it identifies Registered Users within the notification radius of the fire's location.
3. A push notification is sent to each affected user's device, alerting them to the new verified fire nearby.
4. Tapping the notification takes the user directly to the Live Fire Map, centered on the reported fire.



13.2 Viewing Notifications

In addition to push alerts, Registered Users can review past notifications from within the app at any time by clicking the bell icon. This is useful for checking on fires that may have been reported while the user was offline, or to revisit an earlier alert.



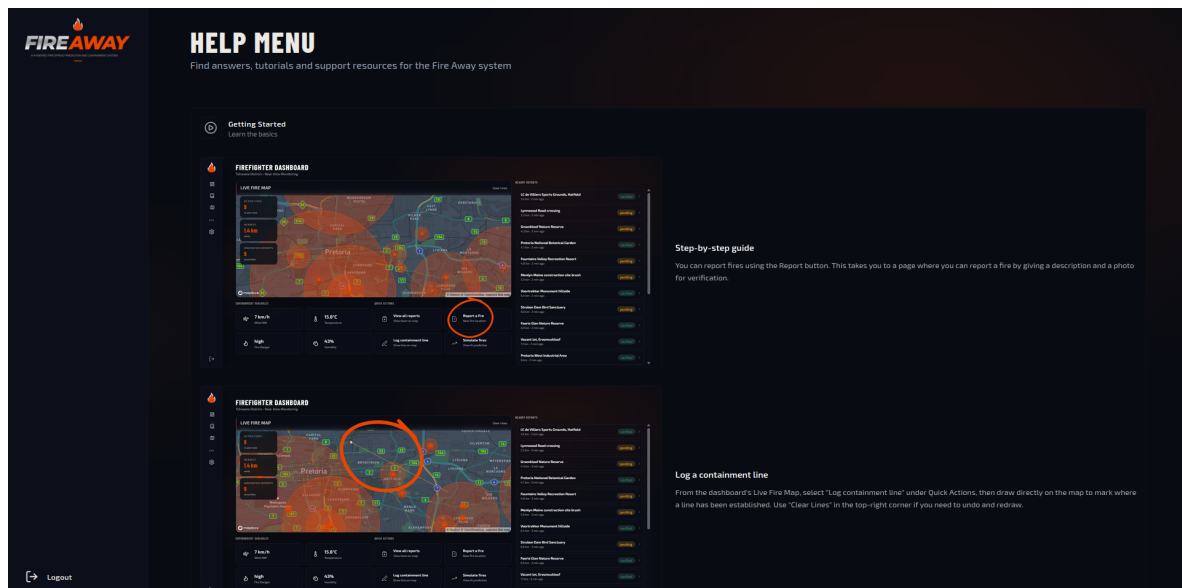
14 Help and Support

The Help Menu can be reached from the sidebar at any time and provides tutorials, frequently asked questions, and emergency contact details.

14.1 Getting Started Tutorials

The **Getting Started** panel is an expandable walkthrough with accompanying screenshots covering the app's core workflows step by step:

- Reporting a fire using the Report button, including a description and photo for verification.
- Logging a containment line from the Live Fire Map's Quick Actions menu, by drawing directly on the map (with an option to clear and redraw).
- Simulating a containment strategy from the Fire Simulation view, by sketching a proposed containment line onto the predicted spread map.
- Running the simulation to see how a drawn containment line affects the predicted spread area, using the current weather inputs.
- Stepping through the simulation timeline using the slider to view predicted spread from now out to 24 hours ahead.



14.2 Help Center

For support beyond the app itself, the Help Center links out to the official South African Fire and Emergency Services website, where users can find professional support resources.

14.3 Frequently Asked Questions

Below the tutorials, an expandable FAQ list answers common questions about the platform, including: what information is needed to report a fire; how to correct or cancel a mistaken report; account requirements for viewing the map versus reporting a fire; how reports are automatically verified; what happens once a report is verified; how the AI spread prediction works and how reliable it is; how the system behaves when weather data or a live connection is unavailable; and what additional tools are available to firefighters.



National Fire Services Indaba 2024

National Fire Services Indaba 2024

Shack Fires Risks

Safer Stoves Campaign

Veldfires Risks

Fire Services Overview

The Directorate: Fire Services (Dir:FS) is responsible for the administration of Fire Brigade Service Act, 1987 (Act No. 99 of 1987) (FBSA) which is one of the key mandates of the Department of Cooperative Governance....

7th National Fire Safety and Prevention Seminar Concept Note

Knynsa Fires Report

Fire Safety Seminar Report 2016

Safer Stoves Campaign Report

Acts & Regulations E - Book

Safety

Training

Urban Search & Rescue

Gallery

Regulatory Framework

Legislation

Projects



WHAT IF I ACCIDENTALLY REPORT A CONTROLLED AGRICULTURAL BURN OR A FALSE ALARM?

CAN I REPORT A FIRE IF I HAVE NO CELLPHONE SIGNAL OR DATA?

DO I NEED AN ACCOUNT TO VIEW THE MAP OR REPORT A FIRE?

HOW DOES A REPORT GET AUTOMATICALLY VERIFIED?

A report is auto-verified when either (1) two independent reports come in for the same location, or (2) it matches a NASA FIRMS satellite hotspot. Until then, it sits as 'Pending' and no alerts are sent. Firefighters and Administrators can also manually verify or reject a pending report.

WHAT HAPPENS TO A FIRE REPORT ONCE IT'S VERIFIED?

WHAT DATA DOES THE AI USE TO PREDICT HOW A FIRE WILL SPREAD?

HOW RELIABLE ARE THE SPREAD PREDICTIONS?

WHAT HAPPENS IF WIND OR WEATHER DATA ISN'T AVAILABLE WHEN A FIRE IS VERIFIED?

CAN I STILL USE THE APP IF MY DEVICE LOSES SIGNAL IN THE FIELD?

WHAT CAN FIREFIGHTERS DO THAT A REGULAR REGISTERED USER CAN'T?

14.4 Emergency Contacts

For life-threatening emergencies, the Help Menu prominently displays direct-dial emergency contact numbers, including the National Fire Emergency toll-free line (10177) and the City of Cape Town Fire & Rescue number (021 480 7700).

15 Change Log

Date	Version	Change
July 2026	1.0	Initial version (Demo 2)
August 2026	2.0	Reported Fires
September	3.0	Admin Analytics+ Help Menu